Ram Calculus for the Spherical Random Perceptron

Positive Gaussian Recursions, Entropy Endpoints,
and Restricted Variational Certificates

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Abstract

We give a source-specific reformulation of the nonlinear Gaussian recursion and spherical entropy in OpenAI's September 24, 2026 manuscript, using actual Ram Calculus results on full nonlinear variations, normalized kernels, finite local moments, and residual composition. We prove additive finite-hierarchy error budgets, a hierarchy-independent terminal Gaussian L^1 estimate, an exact finite-entropy criterion, and a quantitative endpoint accounting identity. For bounded Lipschitz terminal rewards we derive a variational support cutoff from the source's first variation, sharpen its elementary path-continuity constant, and give a finite-cover enclosure theorem and admissible-direction exclusion tests. Bounded Borel rewards are covered by the finite recursion estimates, not by an asserted extension of the source's Brownian control theorem. The source's replica, contact-comparison, and cavity arguments are traced but remain separate from these local certificates. No global proof certification, historical novelty, or computational speedup is claimed. Contribution assessment: C2/5, explicit source-specific error budgets and restricted variational consequences are proved, but the main pressure formula and its probabilistic bridges are not improved or independently certified.

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1 Source, Scope, and Mathematical Contract

The assigned source is *The free energy of the spherical random perceptron* [1], at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. The entire 1,579-line `build/paper.tex`, its one-line main wrapper, proof-map input, bibliography, and README were read. This is mathematical source reading, not an independent reconstruction of every imported theorem. The existing upstream audit reports the selected manuscripts unchanged at `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`; that identity audit is reused, not rerun. Original sources were not edited.

The catalogue assigns applicability L4: the principal analytic functional is directly built from nonlinear Gaussian integrals. L4 does not mean a four-out-of-five contribution, validity assessment, or priority finding. This report's realized contribution is C2 under the project's separate contribution scale. The results below are elementary but useful consequences and quantitative reorganizations, with classical mechanisms identified.

Let $f = \beta\phi$, $\alpha, \beta > 0$, $M_N = \lfloor \alpha N \rfloor$, and let g^a be independent standard Gaussian vectors in $\mathbb{R}^N$. The source uses probability surface measure σ_N on $\{x : \|x\|^2 = N\}$ and

$$p_N[f] = \frac{1}{N} \log \int \exp \left\{ \sum_{a=1}^{M_N} f(g^a \cdot x/\sqrt{N}) \right\} \mathrm{d}\sigma_N(x). \tag{1.1}$$

Pressure has a positive sign; physical free energy is $-p_N/\beta$. For $\mathcal{U}$ the measurable nondecreasing $m : [0,1] \to [0,1]$, put

$$D_m(t) = \int_t^1 m(s)\,\mathrm{d}s, \qquad S(m) = \frac{1}{2} \int_0^1 \left(\frac{1}{D_m(t)} - \frac{1}{1-t} \right) \mathrm{d}t, \tag{1.2}$$

$$V_f(m) = \sup_v \mathbb{E} \left[f\left(B_1 + \int_0^1 m(t) v_t \,\mathrm{d}t \right) - \frac{1}{2} \int_0^1 m(t) v_t^2 \,\mathrm{d}t \right]. \tag{1.3}$$

The controls are progressively measurable with finite expected displayed cost. The entropy is a nonnegative extended integral, with $1/0 = +\infty$. The source's `thm:main` claims, for every $f \in C_b(\mathbb{R})$,

$$\mathbb{E}p_N[f] \longrightarrow \inf_{m \in \mathcal{U}} \{\alpha V_f(m) + S(m)\}, \qquad p_N[f] \longrightarrow \text{the same value in probability.} \tag{1.4}$$

We refer to this as the *source theorem*; our functional estimates do not constitute a new proof of (1.4).

If $a \le f \le b$, zero control and nonnegative cost give $a \le V_f \le b$. Also $m \equiv 1$ has $S = 0$ and $V_f = \log \mathbb{E} e^{f(G)}$. Thus the variational infimum is finite. At finite N, Jensen gives

$$(M_N/N)\mathbb{E}f(G) \le \mathbb{E}p_N[f] \le (M_N/N) \log \mathbb{E} e^{f(G)}.$$

For $f \equiv c$ both sides equal $(M_N/N)c$. These elementary checks fix the normalization; neither involves a replica limit.

2 The Source Proof and Its Unreplaced Inputs

2.1 Quantiles and actual replicas

Represent m as the distribution function of a probability on $[0,1]$, up to null sets and a possible atom at one. Let $q : (0,1) \to [0,1]$ be its nondecreasing quantile. Then

$$D_q(t) = 1 - t - \int_0^1 (q(u) - t)_+ \,\mathrm{d}u. \tag{2.1}$$

For a finite quantile with values $q_0 \le \cdots \le q_{k-1}$ and interval weights $w_i > 0$, $\sum w_i = 1$, set $\zeta_i = \sum_{j<i} w_j$, $q_{-1} = 0$, $q_k = 1$. Thus $\zeta_0 = 0$, $\zeta_k = 1$. The source's recursion is

$$\mathcal{T}_{s,d}g(x) = \begin{cases} d^{-1}\log \mathbb{E}e^{dg(x+\sqrt{s}G)}, & d > 0, \\ \mathbb{E}g(x+\sqrt{s}G), & d = 0, \end{cases} \tag{2.2}$$

$$U_k = f, \qquad U_{i-1} = \mathcal{T}_{q_i - q_{i-1}, \zeta_i} U_i \quad (i = k, \dots, 0), \qquad V_f(q) = U_{-1}(0). \tag{2.3}$$

The terminal residual is the $d = 1$ step; the root is the $d = 0$ step. At $s = 0$ every $\mathcal{T}_{0,d}$ is the identity. None of these endpoints may be replaced by a positive-variance, positive-exponent formula by division.

The Ruelle cascade shared Gaussian field has off-diagonal covariance $q_{\gamma\wedge\eta}$. Each replica visit has a separate residual of variance $1 - q_{k-1}$, *even if two visits select the same leaf.* Consequently a repeated leaf does not make the total marks identical. For example, two visits with shared variance q have covariance q and individual variance one; sharing the residual would incorrectly give covariance one. This distinction is essential for the source's two-pattern second-moment calculation, not just an implementation convention.

For smooth bounded terminal data with bounded derivatives, the heat transform on each time interval gives

$$U_t + \tfrac{1}{2}U_{xx} + \tfrac{1}{2}m(t)U_x^2 = 0, \qquad U(1,x) = f(x). \tag{2.4}$$

Itô's formula along $\mathrm{d}X_t = \mathrm{d}B_t + m(t)v_t\,\mathrm{d}t$ gives exactly

$$\mathbb{E}\left[f(X_1) - \tfrac{1}{2}\int_0^1 mv^2\right] = U(0,0) - \tfrac{1}{2}\mathbb{E}\int_0^1 m(v - U_x(t,X_t))^2.$$

The bounded Lipschitz feedback $v = U_x$ attains equality. This rederives the finite recursion/control bridge; the general continuous-terminal and general-m passages need approximation as in the source. The stochastic control background is classical [9, 8].

2.2 The two bounds

The following account records what was read, and where our certificates stop.

Source location	Actual proof obligation and status here
`sec:prelim`	GG identities, Gram-array representation, support and ultrametricity give finite cascade laws. The source proves the stable-Poisson partition and marked-kernel transformations. These are not consequences of finite local moments.
`lem:fresh-field-passage`	Approximate $\log s$ and s^{-n} uniformly on a positive compact interval. Each polynomial term uses finitely many base replicas and Gaussian covariance continuity, including degeneracies. Bounds on the factor away from zero are indispensable.
`sec:value`	The tilted chain identifies V and its tail-integral order. Finite quantiles rounded downward approximate general trials, with monotone convergence of entropy. Our Sections 3–6 quantify parts of these steps.

Source location	Actual proof obligation and status here
`prop:spherical`	A canonical Gaussian quadratic integral requires every recursive precision positive. Fixed-cascade log-total fluctuations are controlled, then a compact finite-net argument provides uniformity for quantiles bounded by $B < 1$. No depth-uniform fluctuation constant is asserted or needed.
`sec:upper`	Poissonize density and enrich by cascade linear fields. A contact minimum selects perturbation parameters. Concentration plus convex difference quotients enforce joint GG identities there. Synchronization orders spin and label overlaps; tail order contradicts the contact time inequality. This nonlinear-pattern argument is not replaced by a Gaussian mixed-spin formula.
`prop:bulk`	Remove two patterns and restore independent marked components. First and second moments identify $B_{12} = a(R_{12})$ and constant diagonal/correction observables. Conditional independence is with respect to the entire sampled tree shape.
`sec:cavity`	First take $N \to \infty$ at fixed cavity size L, cap the unbounded spherical factor, pass finite-replica limits, then uncap. Average common perturbation parameters, telescope increments, and finally take $L \to \infty$. Finite recursion errors do not justify interchanging these limits.
`sec:completion`	One-pattern deletion controls microscopic terminal approximation. Bounded control cost gives a second-moment tail bound for continuous rewards. Bounded differences gives variance $O(N^{-1})$. Our terminal estimate sharpens an approximation interface, not the microscopic comparison proof.

2.3 Spherical identities and a sensitive boundary

The bulk observables actually used are

$$B_N(x,x') = \frac{1}{N}\sum_a f'(b_a(x))f'(b_a(x')), \quad C_N(x) = \frac{1}{N}\sum_a \{f''(b_a(x)) - b_a(x)f'(b_a(x))\}.$$

For compactly supported smooth f these are bounded. The source proves

$$a(r) = A_q(r) := \int_0^r D_q(t)^{-2}\,\mathrm{d}t \quad (\mu\text{-a.e.}), \quad a \le \alpha \left\|f'\right\|_\infty^2, \quad c + a_{\mathrm{d}} = \int r a(r)\mu(\,\mathrm{d}r). \tag{2.5}$$

The tangential divergence identity has direction $x^2 - R_{12}x^1$ and divergence $-(N-1)R_{12}$. After Gaussian integration by parts and GG conditioning it becomes $a(r)D_q(r) = U(r)$ on the overlap law, where

$$U(r) = r\left(1 + \int (1-s)a(s)\mu(\,\mathrm{d}s)\right) - \int (r-s)_+ a(s)\mu(\,\mathrm{d}s).$$

In particular $U(1) = 1$. Continuity and bounded a exclude support near one. The distributional equations $D_q'' = -\mu$, $U'' = -a\mu$ give $\mathrm{d}(U'D_q - UD_q') = (U - aD_q)\mu = 0$, including interior atoms. Above the support the constant is one, so $(U/D_q)' = D_q^{-2}$ and $U(0) = 0$ proves (2.5). This explains exactly why an endpoint gap holds for the *smooth-model bulk law*. It does not authorize an endpoint gap for every trial in (1.2).

No definite source gap is asserted from this reading. In particular the fresh residual, fractional-moment, and endpoint issues above are treated explicitly in the source. The overlap representation and ultrametricity inputs [11] were not independently reproved, nor were formal proof artifacts or all cited literature audited. Finite diagnostics below do not certify the contact or cavity arguments.

3 Actual Ram Passages and a Positive Recursion

3.1 What is transferred

Three maps were read as routing aids, then the actual following proofs were read directly. Paths and hashes are in the evidence ledger.

1. Astra TR6 [2], *Full Urysohn variation*, `eq:du`, `eq:ddu`, `eq:newtonfull`: for $\mathcal{U}g(x) = \int H(x, y, g(y), g(x))\,\mathrm{d}y$,
$$D\mathcal{U}(g)v = \int (H_s v(y) + H_t v(x))\,\mathrm{d}y.$$
The second derivative has all four source/observation pairings. The proof is the state mean-value formula plus dominated integration. The unbounded-domain subsection requires integrable majorants; Gaussian support is not made compact by the coordinate substitution $y = x - u$. Its normalized-kernel formula `eq:attentiond` retains the derivative of the denominator, giving a centered covariance term.

2. Volume 1 revision 2, Chapter 3 [3], `thm:ram-local`, `eq:ram-integral-remainder`, `eq:ram-remainder-bound`: finite Taylor expansion of the *composed integrand* has a remainder controlled on every contributing segment by an absolute moment. `thm:degree-closure` distinguishes exactness on polynomials from invariant finite degree. Exponentiating a polynomial generally destroys polynomial closure.

3. Astra TR1, `stability.tex` [4], `eq:comp`: errors telescope through Lipschitz maps provided both intermediate states stay in their declared sets. Its `thm:ball/eq:radii` requires a genuine isomorphism, an invariant ball and a contraction margin. TR6's *Ball-local Urysohn solvability* has the same indispensable margin. They do not provide an automatic optimizer or an inverse for the present recursion.

For bounded Borel g, Gaussian measure is finite and e^{dg} is bounded above and bounded away from zero, so differentiation in bounded state directions is justified on the Banach space of bounded Borel functions with its pointwise supremum norm. No spatial smoothness is needed here. With
$$\pi^{s,d}_{g,x}(\,\mathrm{d}y) = \frac{e^{dg(y)}\gamma_{x,s}(\,\mathrm{d}y)}{\int e^{dg}\,\mathrm{d}\gamma_{x,s}},$$
the full nonlinear derivative calculation gives, also at $s = 0$,
$$D\mathcal{T}_{s,d}(g)v = \mathbb{E}_\pi v, \qquad D^2\mathcal{T}_{s,d}(g)[v, w] = d\,\mathrm{Cov}_\pi(v, w). \tag{3.1}$$
For $d = 0$ these mean $P_s v$ and zero. Indeed differentiate the exponential integral once and then the normalized quotient once; the subtracted product of expectations is precisely the denominator derivative. Bounded exponential derivatives dominate the remainders in the supremum norm. Cauchy–Schwarz and $\mathrm{Var}(v) \le \|v\|^2_\infty$ also give $\|D^2\mathcal{T}[v, w]\|_\infty \le d\,\|v\|_\infty\,\|w\|_\infty$.

Theorem 3.1 (Additive recursion residuals)**.** *Each $\mathcal{T}_{s,d}$, for $s \geq 0$ and $0 \leq d \leq 1$, preserves constants and order, maps $[a,b]$ into $[a,b]$, and is one-Lipschitz in the supremum norm. Let bounded approximate functions $\widehat{U}_i$ satisfy*

$$\left\|\widehat{U}_k - f\right\|_\infty \leq \epsilon_{\text{term}}, \qquad \left\|\widehat{U}_{i-1} - \mathcal{T}_{s_i,d_i}\widehat{U}_i\right\|_\infty \leq \epsilon_i.$$

Then

$$|\widehat{U}_{-1}(0) - V_f(q)| \leq \epsilon_{\text{term}} + \sum_{i=0}^{k} \epsilon_i. \tag{3.2}$$

There is no multiplicative hierarchy-depth constant; a depth-independent total still requires a depth-independent sum of local budgets.

Proof. Order and constant translation follow directly from (2.2). If $\|g-h\|_\infty \leq \epsilon$, then $h-\epsilon \leq g \leq h+\epsilon$, giving the Lipschitz claim, including $d=0$. Constant bounds give range preservation. Subtract each exact recursion from its approximate counterpart and use this Lipschitz bound. Backward induction gives (3.2). This is the TR1 composition argument with every exact-map Lipschitz factor equal to one. □

3.2 Positive logarithm certificates and finite Gaussian moments

For $d > 0$ let $M = P_s(e^{dg})(x)$. If $|\widehat{M} - M| \leq \eta$ and $\widehat{M} > \eta$, monotonicity of the logarithm gives

$$\mathcal{T}_{s,d}g(x) \in \frac{1}{d}[\log(\widehat{M}-\eta), \log(\widehat{M}+\eta)], \quad \left|\frac{\log\widehat{M} - \log M}{d}\right| \leq \frac{\eta}{d(\widehat{M}-\eta)}. \tag{3.3}$$

The positivity margin must be proved; a signed Taylor approximation to a positive integrand need not itself be positive. The apparent $1/d$ loss can be avoided near zero as follows.

Proposition 3.2 (Centered small-exponent evaluation)**.** *Suppose $|g-c| \leq H$, and set $h = g-c$ and $k_d(h) = (e^{dh}-1)/d$, with $k_0(h) = h$. Put $A = P_s k_d(h)(x)$. Then*

$$\mathcal{T}_{s,d}g(x) = c + d^{-1}\log(1+dA), \qquad 1 + dA \geq e^{-dH}.$$

If $|\widehat{A} - A| \leq \eta$ and $d\eta < e^{-dH}$, its centered numerical value has error at most

$$\frac{\eta}{e^{-dH} - d\eta}. \tag{3.4}$$

At $d = 0$ use $c + \widehat{A}$ with error η.

Proof. Factor e^{dc} from $P_s e^{dg}$. The derivative in A of $d^{-1}\log(1+dA)$ is $(1+dA)^{-1}$. Every point of the segment between A and $\widehat{A}$ has denominator at least $e^{-dH} - d\eta$. The ordinary mean-value bound proves (3.4). □

For an actual local-moment rule, take a finite radius R, $s > 0$, and an integrand h equal to e^{dg}, $k_d(g-c)$, or g in the linear case. Assume $h \in C^{n+1}$ on $[x-R, x+R]$, $|h^{(n+1)}| \leq B$ there, and $|h| \leq H_0$ outside as well. Taylor's theorem from [3] gives the rigorous rule

$$Q_n(x) = \sum_{j=0}^{n} \frac{h^{(j)}(x)}{j!}\mu_j(s,R), \qquad \mu_j(s,R) = \int_{-R}^{R} u^j \gamma_s(u)\,\mathrm{d}u, \tag{3.5}$$

$$|P_s h(x) - Q_n(x)| \leq \frac{B}{(n+1)!}\int_{-R}^{R} |u|^{n+1}\gamma_s(u)\,\mathrm{d}u + H_0\mathbb{P}(|\sqrt{s}G| > R). \tag{3.6}$$

This follows by integrating the finite Taylor remainder inside the radius and bounding the entire discarded integral. It is not an infinite series claim. Odd moments vanish, and integration by parts gives

$$\mu_0 = 2\Phi(R/\sqrt{s}) - 1, \qquad \mu_{2j} = (2j-1)s\mu_{2j-2} - 2sR^{2j-1}\gamma_s(R). \tag{3.7}$$

Certified enclosures for Φ, the exponential, derivatives, and arithmetic must be added to (3.6). Local observation intervals require uniform derivative and tail bounds over those intervals, plus coverage of all source arguments reached by the preceding level. A mesh-only residual does not satisfy Theorem 3.1. No full numerical solver is claimed in this report.

3.3 The unbounded canonical moment has a different domain

For $X = c + \|z + \sqrt{v}G\|^2/(2b)$, $G \in \mathbb{R}^n$ standard, $b > 0$, and exponent $d > 0$, completion of the square gives

$$d^{-1}\log \mathbb{E}e^{dX} = c + \frac{n}{2d}\log\frac{b}{b-dv} + \frac{\|z\|^2}{2(b-dv)}, \qquad b - dv > 0. \tag{3.8}$$

For $v > 0$, the integral diverges when $b - dv \le 0$; at equality the quadratic decay disappears and the resulting exponential of an affine function has infinite integral. For $v = 0$ the expression is deterministic. The source uses $v = 2(h_i - h_{i-1})$, $d = \zeta_i$, and $b_{i-1} = b_i - dv$. It needs these fractional moments, not the first moment of the full leaf exponential. For example $b = v = 1, d = 1/2$ has a finite fractional moment while exponent one is at divergence. Our bounded range and terminal-density theorems below do not apply to this quadratic canonical factor. Positive precisions and radius concentration remain part of `prop:spherical`.

4 Terminal Approximation Uniform in Hierarchy

Theorem 4.1 (A single range factor for the whole tilted chain)**.** *Let $a \le f, g \le b$ be bounded Borel functions, $W = b - a$, and use any finite source hierarchy* (2.3)*. Then*

$$|V_f(q) - V_g(q)| \le e^W \mathbb{E}|f(G) - g(G)|. \tag{4.1}$$

For bounded continuous f, g this bound also holds uniformly over all $m \in \mathcal{U}$ for the control values (1.3)*. Consequently*

$$\sup_m |V_f(m) - V_g(m)| \le e^W\{\epsilon_R + W\mathbb{P}(|G| > R)\}, \qquad \epsilon_R = \sup_{|x|\le R}|f(x) - g(x)|. \tag{4.2}$$

The finite-hierarchy assertion includes degenerate variances and both exponent endpoints. It is not a Borel extension theorem for the control problem on arbitrary m.

Proof. Interpolate $f_\theta = (1-\theta)f + \theta g$. On the unweighted chain $Z_{-1} = 0$, $Z_i - Z_{i-1} \sim N(0, q_i - q_{i-1})$ independently, so $Z_k \sim N(0,1)$. Differentiate the recursion using (3.1). Its derivative is the expectation of $(g - f)(Z_k)$ under the successive normalized kernels with density

$$L_\theta = \prod_{i=0}^{k} \exp\{\zeta_i(U_i(Z_i) - U_{i-1}(Z_{i-1}))\}.$$

Each kernel integrates to one, including the root with $\zeta_0 = 0$. Telescoping the exponent, rather than multiplying separate worst-case bounds, yields

$$\log L_\theta = f_\theta(Z_k) - \sum_{i=0}^{k-1} w_i U_i(Z_i). \tag{4.3}$$

Every U_i lies in $[a,b]$, hence $e^{-W} \le L_\theta \le e^W$. Integrating the derivative in θ proves (4.1) without spatial regularity. This also justifies the Borel assertion directly, without a continuity approximation of a Gaussian covariance matrix.

For bounded Lipschitz terminals the source's control comparison gives continuity in m in L^1: clip controls at the terminal Lipschitz constant, compare the same controls, and bound reward and cost changes. Finite-step m approximations therefore extend (4.1) to every m. For general continuous bounded terminals use smooth bounded compact-on-compact approximations with a common range. To justify uniform control convergence independently of the desired bound, if $|f|, |g| \le K$, restrict to cost at most $K_1 = 4K + 2$; larger costs lose to zero control. Then $\mathbb{E}|B_1 + \int mv|^2 \le 2 + 2K_1$, so compact convergence gives

$$\sup_m |V_f(m) - V_g(m)| \le \epsilon_R + 4K(1 + K_1)/R^2.$$

This is the source's separate approximation argument. Applying the already proved Lipschitz-terminal version to smooth approximations and passing to their uniform control limits proves (4.1) for continuous data. Finally $|f - g| \le \epsilon_R$ on the compact interval and $|f - g| \le W$ off it, proving (4.2). □

The estimate uses a Gaussian tail, unlike the source's direct second-moment control tail. Its constant grows with the terminal range, so it need not be numerically better at low temperature. Gaussian-tail decay does not imply all admissible controlled terminal laws have this density bound: (4.3) concerns the optimal finite tilted chain.

Proposition 4.2 (Microscopic and variational terminal budgets)**.** *For bounded Borel $f, g \in [a,b]$, at every N,*

$$|\mathbb{E}p_N[f] - \mathbb{E}p_N[g]| \le \alpha e^W \mathbb{E}|f(G) - g(G)|.$$

For bounded continuous f, g the same right side bounds the difference between their variational infima $\inf_m(\alpha V + S)$.

Proof. Differentiate the finite-volume interpolation of f and g. Delete the pattern contributing a chosen summand; its restored factor lies in $[e^a, e^b]$, hence its normalized tilt is at most e^W. Before restoration the pattern is independent of the spin, and its response is standard normal. Sum M_N terms and divide by N. Boundedness justifies all derivatives and expectations. For the variational infimum use Theorem 4.1, multiply by α, and compare each common trial; S is unchanged, including at infinite-entropy trials. □

5 Entropy Without a Spurious Endpoint Restriction

Theorem 5.1 (Exact finite-entropy criterion)**.** *For $m \in \mathcal{U}$ define $\delta = 1 - m$, $R(t) = \int_t^1 \delta(s)\,\mathrm{d}s$. Then*

$$S(m) < \infty \quad \Longleftrightarrow \quad \int_0^1 \frac{1 - m(s)}{1 - s}\,\mathrm{d}s < \infty. \tag{5.1}$$

In particular finite entropy forces $m(1-) = 1$ but does not force a terminal interval on which $m = 1$.

Proof. Since $D = (1 - t) - R$, the entropy integrand is

$$\frac{1}{D} - \frac{1}{1 - t} = \frac{R(t)}{(1 - t)D(t)}. \tag{5.2}$$

If $m(1-) = \ell < 1$, then $D(t) \le \ell(1-t)$ and the entropy is infinite (including $\ell = 0$). If $m(1-) = 1$, choose $t_0 < 1$ and $c > 0$ with $m \ge c$ on $[t_0, 1)$. There $c(1-t) \le D(t) \le 1-t$; before t_0, $D(t) \ge D(t_0) > 0$, so the integral is finite on that compact part. By Tonelli,

$$\int_{t_0}^{1} \frac{R(t)}{(1-t)^2}\,\mathrm{d}t = \int_{t_0}^{1} \delta(s)\left(\frac{1}{1-s} - \frac{1}{1-t_0}\right)\mathrm{d}s. \tag{5.3}$$

Comparison by factors 1 and $1/c$ in (5.2) therefore gives (5.1). Conversely, finiteness of the right side of (5.1) forces $\delta(1-) = 0$ by monotonicity, and the same argument applies. □

For $p > 0$ and $0 < c \le 1$, the nonplateau trial $m(t) = 1 - c(1-t)^p$ has

$$D(t) = (1-t)\left(1 - \frac{c(1-t)^p}{p+1}\right), \qquad S(m) = \frac{1}{2p}\log\frac{p+1}{p+1-c}. \tag{5.4}$$

Substitute $z = 1-t$ and then z^p in (5.2) to obtain the displayed integral exactly. In particular $m(t) = t$ has $S = \frac{1}{2}\log 2$. Conversely, $1 - m(t) = 1/\log(e/(1-t))$ tends to zero but violates (5.1); hence $S = +\infty$. The condition $m(1-) = 1$ alone is insufficient. Constant $m = c < 1$ also has infinite entropy.

Proposition 5.2 (Full endpoint replacement cost)**.** *Let $S(m) < \infty$, $b < 1$, and let $m^b = m$ below b and $m^b = 1$ above b. Write $e_b = \int_b^1 (1-m)$ and $S_{\mathrm{tail}}(m;b) = \frac{1}{2}\int_b^1 (D^{-1} - (1-t)^{-1})\,\mathrm{d}t$. Then*

$$S(m) - S(m^b) = \frac{1}{2}\int_0^b \frac{e_b}{D(t)(D(t)+e_b)}\,\mathrm{d}t + S_{\mathrm{tail}}(m;b). \tag{5.5}$$

Both terms are nonnegative. The first is at most $\frac{1}{2}e_b\int_0^b D(t)^{-2}\,\mathrm{d}t$.

Proof. For $t \le b$, $D_{m^b}(t) = D_m(t) + e_b$; for $t > b$ it is $1-t$. Subtract the two entropy integrals. Finite entropy makes the subtraction legitimate. The bound follows by enlarging the integrand. □

Changing the endpoint changes D on the whole preceding interval; retaining just the entropy tail is wrong. If $1 - m(t) \le C(1-t)^p$ on all of $[0,1)$ with $p > 0$ and $d_0 = 1 - C/(p+1) > 0$, then

$$S(m) - S(m^b) \le \frac{Cb(1-b)^p}{2(p+1)d_0^2} + \frac{C(1-b)^p}{2p(p+1)d_0}. \tag{5.6}$$

Indeed $D \ge d_0(1-t)$, $e_b \le C(1-b)^{p+1}/(p+1)$, and $R(t) \le C(1-t)^{p+1}/(p+1)$ in (5.2). These bounds prove (5.6) by integration. The constants depend on a declared endpoint modulus; no universal truncation rate over all finite-entropy trials is asserted.

6 First Variations and a Justified Restricted Domain

Lemma 6.1 (Source derivative with its exact normalization)**.** *For a finite quantile and smooth bounded f with bounded derivatives, let Z_i be the tilted chain in the proof of Theorem 4.1 and set $c_i = \mathbb{E}[U_i'(Z_i)^2]$. Then*

$$\dot{V}_f(q) = -\frac{1}{2}\sum_{i=0}^{k-1} w_i \dot{q}_i c_i, \qquad \dot{S}(q) = \frac{1}{2}\sum_{i=0}^{k-1} w_i \dot{q}_i A_q(q_i), \tag{6.1}$$

where the entropy formula requires $q_{k-1} < 1$. Moreover $0 \le c_0 \le \cdots \le c_{k-1} \le L^2$ if $L = \|f'\|_\infty$.

Proof. Differentiate (2.2) spatially: $U'_{i-1}(Z_{i-1})$ is the conditional expectation of $U'_i(Z_i)$. Thus the derivatives form the bounded martingale with terminal $f'(Z_k)$, proving the assertions on its second moments. Varying a time q_i lengthens the preceding interval and shortens the following one. The heat identity inserts $\frac{1}{2}(U''_i + \zeta_i(U'_i)^2)$ and subtracts $\frac{1}{2}(U''_i + \zeta_{i+1}(U'_i)^2)$. The second derivatives cancel; the remaining term is $-w_i(U'_i)^2/2$. Earlier operators differentiate by tilted expectation, proving the first formula.

From (2.1), $\partial_{q_i} D_q(t) = -w_i \mathbf{1}_{t<q_i}$ for almost every t. Near a finite quantile with top below one, the entropy integrand is identically zero near one and denominators on the remaining compact interval have a positive lower bound. Dominated differentiation gives the second formula. At ties and zero-length intervals take one-sided order-preserving variations; alternatively use strictly ordered approximations and the same bounded formulas. □

This is the source's `eq:tail-order` derivative and `prop:spherical` derivative, rederived with the normalized-kernel calculus. It is not credited as a new derivative formula. Summation by parts with c_i increasing recovers the source's tail order: if all tail integrals of $p - q$ are nonnegative, then $V_f(p) \le V_f(q)$. The terminal need not be convex or even.

Theorem 6.2 (Sharpened continuity and Lipschitz-trial cutoff)**.** *Let f be bounded and globally Lipschitz with constant L, and put $C = \alpha L^2$ and $B = C/(1+C)$. For all quantiles p, q,*

$$|V_f(p) - V_f(q)| \le \tfrac{1}{2} L^2 \|p - q\|_1 . \tag{6.2}$$

For the functional $F_f(q) = \alpha V_f(q) + S(q)$ and the cap $q^B = \min(q, B)$,

$$F_f(q^B) \le F_f(q), \qquad \inf_{q:[0,1]\to[0,1]} F_f(q) = \min_{q:[0,1]\to[0,B]} F_f(q), \tag{6.3}$$

where every quantile in the infimum/minimum is nondecreasing, modulo null sets. Thus at least one minimizing trial is equal to one above B. This conclusion is proved for bounded Lipschitz f, not assumed for arbitrary bounded continuous f.

Proof. For smooth bounded f with bounded derivatives, integrate the first formula of (6.1) along the segment joining finite quantiles on common weight intervals. Since $c_i \le L^2$, this gives (6.2). Endpoints follow by continuity. Smooth mollifications of a bounded Lipschitz f preserve the Lipschitz constant and converge uniformly; terminal supremum-norm nonexpansivity passes the inequality to f. The control continuity argument and finite-step density extend it to all quantiles. The equality between distribution-function and quantile L^1 distances may equivalently be obtained by integrating the area between their graphs. The coefficient $1/2$ improves the source's elementary clipped-control coefficient $3/2$ at this interface only.

For a finite quantile with top below one, $D_q(t) \le 1 - t$, whence $A_q(r) \ge \int_0^r (1-t)^{-2}\,\mathrm{d}t = r/(1-r)$. Therefore

$$\frac{2}{w_i}\partial_{q_i} F_f(q) = A_q(q_i) - \alpha c_i \ge \frac{q_i}{1-q_i} - C. \tag{6.4}$$

Lower a common cap from $\max q_i$ to B. Each moving coordinate has $q_i \ge B$, so the derivative along that motion is nonpositive. Tied coordinates move together, preserving order. This proves (6.3) for smooth finite trials and, by uniform mollification, for Lipschitz terminals. Approximate a general finite-entropy q from below by finite quantiles capped strictly below one. Then $S(q_j) \uparrow S(q)$ by (2.1) and monotone convergence, and $V(q_j) \to V(q)$. The capped quantiles converge in L^1 within $[0, B]$; their entropies converge since denominators below B are at least $1 - B$ and above B the integrand vanishes. Passing limits gives the cap inequality. For infinite-entropy q it is automatic.

The nondecreasing $[0, B]$-valued quantiles form a compact set in L^1 by monotone selection and bounded convergence. Both terms of F are continuous there by the estimates just given, so a minimum exists. If $L = 0$, f is constant and $q = 0$ attains the minimum directly. □

The cap equals the source's smooth bulk support bound, but its use here is a functional minimization consequence of the source derivative, not an inference that every admissible trial has bounded support. When approximating a merely continuous f, mollification Lipschitz constants may diverge and B may tend to one. Entropy constants must then be retained; a common strict cap has not been proved.

Proposition 6.3 (Admissible interval derivative test)**.** *Fix positive weights and a compact ordered parameter region with top at most $B < 1$. Suppose rigorous enclosures throughout this region give $A_q(q_i) \in [A_i^-, A_i^+]$ and $c_i(q) \in [c_i^-, c_i^+]$. Then $\partial_{q_i} F$ belongs to*

$$\frac{w_i}{2}[A_i^- - \alpha c_i^+,\ A_i^+ - \alpha c_i^-].$$

If a fixed nonzero direction d is feasible for a short positive step at every point in the region, and the upper bound on $\sum_i d_i \partial_{q_i} F$ is strictly negative, no such point is a local minimizer. At ties, feasibility requires $d_i \le d_{i+1}$ for equal coordinates; at a zero coordinate it requires $d_i \ge 0$. Artificial search caps add their own inward-direction constraint.

Proof. Subtract intervals with the sign of $-\alpha$ retained. For the sum, multiply each interval by d_i, reversing endpoints when $d_i < 0$. The uniform strict negative bound and the fundamental theorem along the feasible segment imply a decrease. The feasibility conditions are exactly the tangent inequalities of the ordered cube. □

For implementation c_i is an expectation of a squared martingale, not a signed covariance guessed from samples. Normalized numerator and denominator interval enclosures must include their common positive denominator margin. The universal $[0, L^2]$ enclosure already proves (6.4); tighter intervals require actual validated integration. No root, uniqueness, or global minimum follows from a sampled gradient.

7 A Finite-Cover Enclosure Theorem

This section states exactly what would be needed to turn the preceding local machinery into a full *functional* enclosure for Lipschitz data. It does not report that such a global computation was performed.

Theorem 7.1 (Finite hierarchy coverage with a complete budget)**.** *Let f be bounded Lipschitz, $L > 0$, and take B as in Theorem 6.2. Let $\mathcal{G}_{k,J}$ consist of all nondecreasing quantiles constant on k equal intervals and taking values in $\{0, B/J, \dots, B\}$. Set*

$$C_F = \frac{\alpha L^2}{2} + \frac{B}{2(1-B)^2}, \qquad \eta = B/k + B/J. \tag{7.1}$$

If every grid functional has an enclosure $F_f(q) \in [\ell_q, u_q]$, then

$$\min_{q \in \mathcal{G}_{k,J}} \ell_q - C_F \eta \le \inf_{m \in \mathcal{U}} (\alpha V_f(m) + S(m)) \le \min_{q \in \mathcal{G}_{k,J}} u_q. \tag{7.2}$$

The grid has $\binom{k+J}{k}$ elements. Certified pruning may remove elements only with a valid covering or exclusion argument.

Proof. If $p, q \le B$, formula (2.1) gives $\|D_p - D_q\|_\infty \le \|p - q\|_1$. On $[0, B]$ both denominators are at least $1 - B$; above B both entropy integrands vanish. Thus

$$|S(p) - S(q)| \le \frac{B}{2(1-B)^2} \|p - q\|_1 .$$

Together with (6.2) this gives the C_F Lipschitz bound. Approximate a monotone $q \in [0, B]$ by its average on each equal interval. Its L^1 error is at most B/k: bound each cell by its oscillation times $1/k$ and sum the oscillations, which are at most total variation B. Rounding these ordered averages down to multiples of B/J adds at most B/J. This constructs a grid point within η. Apply this to a minimizing quantile furnished by Theorem 6.2. It proves the lower inequality; every grid point is feasible, proving the upper one. Counting weakly increasing k-tuples from $J + 1$ values gives the binomial coefficient by stars and bars. □

For each grid trial, an enclosure radius may be assembled as

$$E_q = \alpha \left(\epsilon_{\text{term}} + \sum_i \epsilon_i \right) + \epsilon_S. \tag{7.3}$$

Here ϵ_i must contain moment, tail, derivative, interpolation, roundoff, and positive-log errors in a common supremum norm; ϵ_S encloses the actual entropy integral. The finite step entropy can also be evaluated exactly up to scalar logarithms:

$$2S(q) = \frac{q_0}{D_0} + \sum_{i=1}^{k-1} \frac{1}{\zeta_i} \log \frac{D_{i-1}}{D_i} + \log D_{k-1}, \quad D_i = D_q(q_i),\ D_{k-1} = 1 - q_{k-1}. \tag{7.4}$$

This follows by integrating $1/D$ on each interval where D is affine, and canceling the reference endpoint logarithm. Tied positions give zero log ratios. Positive D_i must remain certified.

For continuous bounded f, choose bounded Lipschitz g in a common range $[a, b]$ and add $\alpha e^{b-a} \mathbb{E}|f(G) - g(G)|$ to both ends of a certificate for g. This yields a valid approximation route with its possibly deteriorating cap and complexity. A finite collection of trial upper bounds alone gives no variational lower bound. To interpret (7.2) as a limit-pressure enclosure one must additionally accept the source theorem (1.4); no finite-N convergence rate is supplied.

8 Verification, Benefits, and Remaining Work

8.1 Reproducible diagnostics and their limits

The accompanying `evidence/checks.py` uses rational arithmetic for finite identities and high-precision arithmetic for numerical diagnostics. Exact tests cover a rational entropy derivative, its positive-precision canonical recursion, endpoint-denominator changes, the tilted-density telescoping identity, and valid/invalid trial cutoffs. Numerical tests compare (5.4) with quadrature, check the closed one-level $d = 1$ terminal case, and check (3.7). They are deterministic and replayable.

Exact arithmetic certifies only the displayed finite rational examples. High precision without directed interval error bounds is diagnostic, not rigorous quadrature certification. These tests do not establish all hierarchies, the global grid minimum, overlap identities, stochastic control existence for Borel terminals, or the source's thermodynamic limit. The theorem proofs above, not observed numerical agreement, support the general estimates.

8.2 Advantages and costs

The positive-kernel formulation has four practical benefits: it preserves the range and both exponent endpoints; it makes local errors additive; it prevents exponentiation from being silently treated as polynomial closure; and it exposes the positivity margin before applying a logarithm. The tilted-chain telescoping gives one terminal range factor, independent of depth. The entropy criterion identifies precisely which nonplateau trials remain legal. For Lipschitz data, the justified cutoff gives a compact quantitative search domain and a finite-cover lower-bound route.

The costs are substantial. Uniform source-domain coverage and derivative bounds are stronger requirements than pointwise quadrature accuracy. e^W can be large, $1 - B$ can be small, and the grid cardinality is combinatorial. Local polynomial coordinates may worsen conditioning; they do not create inverse stability. TR1's nonlinear ball theorem cannot be invoked at constant-shift directions as a strict contraction: the recursion preserves such shifts exactly. No empirical performance comparison or speedup has been measured.

8.3 Assumptions and honest contribution

All recurrence range and residual results need bounded intermediate states. Moment rules need derivatives of the actual composed integrand on complete segments and all Gaussian tails. The terminal L^1 proof uses increasing hierarchy exponents and total Brownian variance one. The full control statement is restricted to bounded continuous rewards. Entropy finiteness is exactly (5.1). The cutoff/cover theorem requires a global Lipschitz constant; it does not apply uniformly to all C_b rewards. Derivative exclusion needs whole-region enclosures and admissible directions. The canonical quadratic formula needs all successive precisions positive.

The C1–C5 scale is separate from L1–L5 applicability: C1 means no contribution demonstrated; C2 a small contribution; C3 notable/significant; C4 substantial/high; C5 very high, substantial and significant improvements. We assess C2, because the explicit endpoint and approximation budgets and the cutoff/cover consequence are useful but obtained from classical positive kernels and source derivatives. The source already has its main formula, qualitative continuity, a smooth bulk support cap, and finite-step density. Those are not credited again as report advances. No historical novelty, source-gap closure, uniqueness of an optimizer, replica-symmetric reduction, or all-temperature algorithmic optimality is claimed.

Future work should first implement genuinely validated moment/log arithmetic with propagated spatial domains, then test adaptive finite coverings against the explicit combinatorial cost. A quantitative finite-cavity error rate would be a different, harder theorem. Borel control identification on general m, unbounded terminal potentials, zero-temperature limits, and hard-constraint/jamming limits require additional integrability and limit arguments and are not corollaries of this report. Independent review of the source's contact selection, two-pattern conditional laws, and cavity uniformity remains worthwhile.

9 Literature and Read-Scope Record

Public searches were limited to public titles, author names, and bibliographic identifiers; no private passages or corpus filenames were used as web queries. The bounded literature check was made on October 8, 2026 and is not an exhaustive novelty review.

Györgyi and Reimann [12] give the historical continuous-RSB prediction; Shcherbina and Tirozzi [13] treat a different rigorous Gardner observable/regime. These are cited as source dependencies, not independently reaudited here. Montanari and Zhou's version 3 record and Appendix B were

checked directly [5]; their finite-temperature replica calculation is explicitly nonrigorous, whereas their algorithmic control result has a different scope. The primary arXiv record says accepted to *Annals of Probability*; no unverified final pagination is supplied. The formula dictionary in the assigned manuscript must retain the per-coordinate factor $\alpha\beta$.

Chen and Mourrat [6] characterize critical values for nonconvex Gaussian vector-spin models subject to existence of the limit, not this non-Gaussian pattern Hamiltonian. Chen and Issa's March 2026 preprint [7], checked at its primary abstract, proves a high-temperature result for nonconvex vector-spin/Mattis models; it does not remove the assigned source's model-specific comparison obligations. Auffinger and Chen [8] establish a strict-convexity/uniqueness result for their Parisi setting. That uniqueness is not imported for arbitrary perceptron terminal rewards here. The control, cascade, and Gaussian integration methods have established antecedents [9, 10, 11].

Additional primary records supplied during parent coordination were checked at their metadata/abstract level: Chen–Issa–Mourrat version 2, dated September 29, 2026, addresses centered Ising multi-species spins [14]; Ho's June 2026 paper concerns concavity with respect to a time-changed Parisi diffusion profile [15]; Shmalo's July 2026 preprint claims a computer-assisted zero-margin Ising storage-capacity result [16]. These are different spin domains, parameterizations, or hard-constraint observables. Their full proofs and certificates were not checked here, and none is used to assert concavity, a hard-constraint limit, or a pressure formula for the spherical bounded-reward problem.

The actual private proof reading was selective: TR6 lines 1–490 and 630–719, including the full variation and ball-local proofs; Volume 1 Chapter 3 lines 1–225, including the complete local Taylor remainder and degree-closure proofs; and the whole Astra TR1 `stability.tex`. Embedded chapter inputs, remaining TR6 sections, all other corpus theorems, and their recorded numerical suites were not read or replayed. The three maps were routing material, not substitutes for those proofs. The source's mathematical body, wrapper, proof map, references, and README were fully read; referenced external books and papers were not all fully read. No Ising-perceptron report was duplicated.

The accompanying reproducibility package records source paths and SHA-256 identities, finite diagnostics, public-literature observations, and compiler and all-page rendering checks. The diagnostic script separates exact rational checks from high-precision numerical tests; neither replaces the proofs or extends their stated domains.

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